FINANCIAL RISK PROFILING ANALYTICS



THESIS SUBMITTED TO

Symbiosis Institute of Geoinformatics

FOR PARTIAL FULFILMENT OF THE M. Sc.

DEGREE

By

ATHARVA DADA HANDE

(Batch 2024-26 / PRN 24070243019)

Symbiosis Institute of Geoinformatics

Symbiosis International (Deemed University)

5th Floor, Atur Centre, Gokhale Cross Road,

Model Colony, Pune – 411016

UNDERTAKING

This is to certify that the project titled Financial Risk Profiling Analytics is a Bonafide work done by Atharva Hande at Symbiosis Institute of Geoinformatics.

Internal Supervisor

Dr Vidya Patkar

Deputy Director at Symbiosis Institute of Geo-informatics

Table Of Content:

1. **Acknowledgement…….. 4**
2. **List of Tables…… 5**
3. **List of Figures……...6**
4. **Introduction……..7 - 8**
5. **Literature Review……9 – 20**
6. **Literature Review Table…….. 21 - 26**
7. **Methodology…….27 - 32**
8. **Result and Analysis……..33 - 41**
9. **Investor Behavioral Recommendation System..42-45**
10. **Limitations ……46**
11. **Drawbacks……..47**
12. **Conclusion and Future Scope……..48 - 49**
13. **References……..50 - 51**

Acknowledgement:-

We would like to express our sincere gratitude to our internal supervisor Dr Vidya Patkar , for her valuable guidance for the entire tenure of our project, guiding me from the start of the object till today. Moreover, the feedbacks which always emphasized more on improvement rather than satisfaction has helped me do the thing that I've been doing since the past three months, The valuable insights from the supervisor has been of great importance, and gave the confidence to do what was needed and greatly enhanced the project, than what was expected from myself.

I would also like to thank the faculty members our institution Symbiosis Institute of Geo-informatics for providing us with the support to complete the project in the desired way, finally I greet my family for all the love and support which included constant support and motivation which helped me in the entire life-span of my project.

Atharva Dada Hande

MSc Data Science and Spatial Analytics

PRN:-24070243019

List of Tables:-

SR.NO	Table Name
1.	Traditional Finance Vs Behavioral Finance
2.	AI Techniques for Behavioural Risk Management
3.	Investor Profiling
4.	Significance of Weights
5.	Count of Risk Types
6.	Recommendation Count

List of Figures: -

SR.NO	Figure Name
1	Analysing Customer Behaviour
2	Breaches(21-26)
3	Breaches (16-21)
4	Threat Detection Techniques
5	Factors Influencing FRT
6	Clustering and Decision Tree
7	Cluster Summary (DB-Scan)
8	Cluster Summary (K-Means)
9	Silhouette Score (DB-Scan)
10	Silhouette Score (K-Means)
11	Distribution of risk categories
12	Motto for risk
13	Gender of risk takers
14	Risk Vs Investment
15	Distribution of recommendation count
16	Recommendation Vs Risk Count

Introduction

Investment decisions are not only influenced by income or financial capability but are majorly influenced by an individual's behavior, understanding of risk, and personal habits of handling or using money as an asset. And thus, understanding a person's financial habit has become an important area of research in this dynamic world, moreover, having control over finances is dependent majorly on having control over personal emotions, unrealistic expectations, personal life traumas and many more pure human emotions.

The financial sector has undergone a significant transformation with the increasing adoption of data-driven technologies and intelligent analytical methods. Traditional investment decisions were primarily guided by financial advisors, personal experience, and an individual's perception of market conditions. While these approaches continue to play an important role, the growing availability of financial data has created opportunities to understand investor behaviour more systematically through analytical techniques. As a result, organizations are increasingly relying on data analytics and machine learning to improve investment decision-making, customer profiling, and financial planning.

Investor behaviour is influenced by a combination of financial objectives, personal preferences, and psychological factors. Individuals with similar financial goals may still demonstrate different investment patterns because of variations in their willingness to take risks, preference for stable investment options, portfolio diversification, and the frequency with which they monitor their investments. These behavioural differences make investor profiling a challenging task when performed solely through conventional methods, which often depend on subjective judgement and manual assessment.

Our project mainly focuses on analyzing how an individual behaves while making personal investment decisions and also term each individual upon the risk associated with each investment as such, also the dataset used in the project will consist of investment preferences such as mutual funds, fixed deposits, PPF, investment duration, stock market, gold, government bonds by which thorough analysis of investment habits can be done

The main primary focus is instead of predicting financial with a very basic statistic like the credit score we are going forward and identifying risk related to a person's behavior various key variables are to be derived like risk appetite score, investment stability, diversification behavior, monitoring frequency would prove to be the key factors for our analysis and by which we could build a model that can classify the spenders based on risk like low risk, moderate risk and high risk.

Objectives of the study

The prime objectives of the project are:

1) To analyze individual behavior using basic behavioral financial concepts

2) To develop a model based on indicators such as risk appetite score, stability index, diversification index and monitoring intensity score.
3) Understanding the influence of behavioral study with respect to financial risk on investments
4) To classify investors as low risk, high risk, moderate risk based on the pattern indicated by our decision variables

Methodology

The methodology of this project focuses on analysing investor behaviour to understand different levels of financial risk tolerance. The dataset contains information about individuals' investment preferences such as mutual funds, fixed deposits, PPF, investment duration, and other behavioural indicators. The first step in the process will be **data cleaning and preparation**, where the dataset will be reviewed to remove inconsistencies, handle missing values, and ensure that all variables are structured properly for analysis. After preparing the dataset, behavioural indicators such as **Risk Appetite Score, Stability Index, Diversification Index, and Monitoring Intensity Score** will be developed using relevant variables from the dataset. These indicators will help capture different aspects of how individuals approach financial investments and risk-taking behaviour.

Instead of using a manual scoring system to categorize investors, this study will apply a **K-Means clustering technique**, which is a data-driven method used to group individuals with similar characteristics. The clustering algorithm will analyse patterns within the behavioural indicators and investment preferences to identify natural groupings among investors. Based on the similarities in their financial behaviour, the algorithm will classify individuals into different clusters that represent varying levels of financial risk tolerance.

These clusters will then be interpreted as **Low Risk, Moderate Risk, and High Risk investor categories**. This approach allows the classification to emerge from the behavioural patterns present in the data itself rather than relying on predefined scoring thresholds. By using clustering analysis, the study aims to provide a more objective and analytical understanding of how behavioural factors influence financial risk-taking in investment decisions.

Expected outcome

The expected outcome of this project is to identify different types of investors based on their financial behaviour and investment preferences. By analysing behavioural indicators such as risk appetite, stability preference, diversification behaviour, and monitoring habits, patterns in investment decision-making can be observed. Using the **K-Means clustering technique**, investors will be grouped into clusters based on similarities in their behavioural characteristics. These clusters will then be interpreted as **low-risk, moderate-risk, and high-risk investor profiles**, helping to better understand how behavioural factors influence financial risk-taking.

2) Literature Review

2.1) Risk Profiling/ Behavioral Risk

Risk profiling in financial terms is the process of analyzing a person's approach towards taking risks, in simpler words, we can say that risk profiling is the amount of risk a person can indulge in while investing. Basically, this can answer a few questions like

i) How much risk can a person sustain in order to reach his financial goals?

ii) Is the risk taken worth the reward?

Risk as a factor is always associated with reward, and getting the optimal reward requires a lot of financial planning and experience, similarly if we talk about investments a person's behavior matters more than the knowledge about investments, impulsive behavior ultimately shows up irrespective of knowledge about a particular domain, mostly investments, so understanding a person's behavioral approach is more important than the on paper credit scores, previous investment history etc.

According to MR Salem Kattoura, DR Luk Chang Leung and DR Patrick Leung (2022)

As cited in [Investor Risk Profiling: An Enhanced Behavioural Finance Perspective and Technique for Identifying Client Risk Profiles] managers are starting to understand the depth of communicating crucial information, where from prior experience's they have identified the importance of uplifting of the managerial process by applying behavioural theories.

However the effect of behavioural approaches and information sharing can vary with development, like the free market situation in developed countries makes it easier to spread information faster because of its financial and infrastructure equality, However, in developing countries uneven information spread is very common due to uneven development, uneven spread of resources which affects the information flow, access to information. (Ishii and Habermaier, 2002). Risk profiling is involved with the character and the capacity to stretch a person's limitations in order to reach the financial goal, however profiling in this investment and return aspect is involved with behavioural aspects and reaction to risk involved (Kimball, et al., 2008) Although, in the same study by Kimball the literature in psychology provides studies that involve studying psychological traits and clustering similar psychologies in a group, and behavioural risk profiling uses such studies at its core to examine financial risk and risk tolerance of a certain individual, (Kimball, et al., 2008) also underlines the unfair means that clients face while clustering them, as it nullifies their unique traits and can oversimplify their choices based on one particular thing while neglecting the other and can prove to be over generalisation which is hazardous to the private and financial sector

As cited in [Investor Risk Profiling : An Overview] it asks two questions

i) What is a risk profile understood to be?

ii) What are the most common risk profiling technique used by practitioners?

Where risk profiling is defined a mix of classical decision making, modern portfolio theory and capital asset pricing model (CAPM) to define the risk profile of an investor

[Developing Predictive Models for Financial Stability: Integrating Behavioural Analytics into Credit Risk Management (2024), Saugat Nayak] the limitations of traditional financial risk models, the behavioural aspect has turned out to be crucial and important indicator to uplift model accuracy, where real time data is used to analyse individual consumer behaviour which gives us a more dynamic and detailed view of risks where unlike traditional methods which is built on historic variables, this behavioural approach also captures various important variables like spending pattern, social media trends etc ,which traditional models can miss.

Behavioural finance	Traditional Finance
It combines finance and human psychology and often considers investors to be irrational	Investors are considered to be rational. And assumed as investing only for maximum profit
Here decisions are always related with psychology, human intervention .	Here decisions are totally made based on maximum profit.
Decisions are influenced by peer pressure, copying each other's financial trait	Data is the key indicator for decision making
Example:- A person takes a bank loan with higher interest rate because of quicker loan approval and attractive marketing.	Example:- A person usually goes for a loan with optimal interest rate , after comparing the interest rate of multiple banks, and selects the bank which suits his needs.

Table 1: Traditional Finance VS Behavioural Finance

Behavioural risk is basically losses that can occur because of potential human behaviour like judgemental errors, unethical actions basically those which are non-quantifiable through historic-data and financial metrics, as behaviour is fully qualitative which even makes it harder to negotiate and measure the losses in quantities (Artificial Intelligence for Behavioural Risk Analysis in Finance, Santosh Chitraju Gopal Verma(2025)).

Some of the common Behavioural Risk factors can be undermined as

1) Overconfidence Bias
2) Unethical conduct
3) Moral Hazard

Which eventually results in Investment lapses and has an adverse effect on the capital (Gopal Verma,25)

i) Artificial Intelligence in Behavioural Finance :-

AI systems use dynamic learning techniques to detect risk and its patterns, according to Santosh Verma (25) traditional data depends a lot on historical datasets and rules static in nature, whereas monitoring large data to uncover information that are majorly hidden can be done by AI techniques, where basically we term it as proactive risk analysis rather than depending on reactive risk analysis. Although AI being a very broad scope, sub parts of AI include machine learning for behavioural pattern detection, Natural Language processing for Behavioural Insights, Building predictive models that can analyse the amount of risk

ii) Advantages of AI in financial risk management

AI driven techniques use transactional data, customer behaviour AI models can signal at abnormalities in data and can signal risk, According to Saleem Ashraf, Najif Khalid .AI based models are useful as it can continuously analyse data as it becomes available, that is real time data analysis and risk mitigation becomes available due to Machine Learning and Artificial Intelligence, The most important benefit of AI driven risk modelling features is that it runs real time as soon as data is available, which also enhances customer segmentation by analysing different segment of customers that vary in nature and hence it becomes easier to evaluate what each section of the society is more driven towards and hence completes the meaning of customer segmentation, which **i**s closely related to marketing as when the data shows how each customer is different from each other as well as groups each customer according to similarities and hence improve the marketing trends fed to customers.

K-Means clustering	**An unsupervised learning algorithm used to cluster data or in risk management context individuals into clusters that show similar characteristics.**
Principal Component Analysis	**At it's core PCA generally reduces variables and keep the most promising variables that can be used to group or cluster individuals, removing noise and only focusing on what's need is what PCA is.**
Decision Tree	**Most common supervised learning algorithm, works with branches and can helps in understanding how various variables are influenced s (behavioural factors influence risk) , creating a tree like structure**
Random Forest Algorithm	**Various decision trees working together at a time and combining results of each to improve accuracy of the result**

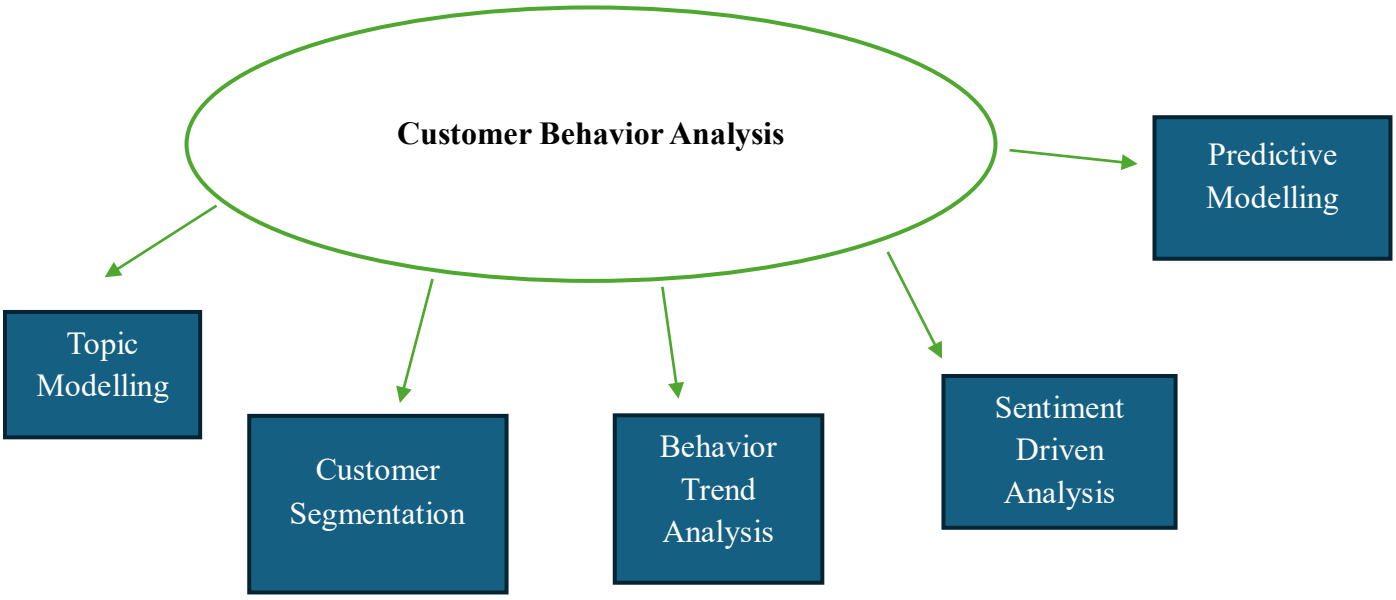
Table 2: AI Techniques for Behavioural Risk Management

2.2) Natural Language Processing for Behavioural Analytics

Natural Language Processing (NLP) is understanding human sentiments from textual data, that is understanding text and speech, extracting meaning from text and responding like a human. Surveys and feedback forms were considered to be the most important form of understanding what the customer thinks, but often fall shorts of understanding the nuance of sentiments spread across social media and other customer interactions (Dr M Suresh , Dr.G.Vincent , Dr.C.Vijai , Dr.M.Rajendhiran , A.H.Vidhyalakshmi , Dr Sundarapandiyan Natarajan, 2024). In the long run behavioural analytics and sentimental analysis or NLP can go hand in hand where NLP can be used to extract any meaningful information or expressions from how the text is presented such as customer interaction reports, loan application etc, also behavioural analytics here helps in analysing the behavioural spending of a person like for example how robust or how frequent a person is for taking a loan, his/her spending habits, approaches towards finance etc.

According to DR Suresh (24) customer behaviour analysis is done in the following ways

Figure 1: Analysing Customer Behaviour



i) Advantage of Behavioural Analysis in Financial Risk Assessment:

1) Enhance identification of risky customers: By analysing the behavioural pattern of customers getting a precise view of risky customers become very easy, many customers can be clean on paper that is customer's with efficient credit score, few or no previous loans or pending delays as such, but this tells us a very short story from a bigger picture but with behavioural risk identification techniques we are able to analyse complete financial behaviour of a person like how much does he spends, how much does he earns, how much does he saves and hence getting a better picture of financial behaviour or financial attitude can be discovered.

2) Easy risk management or institutions: Once, financial behaviour of a person is analysed to identify potential risks, it becomes easy in adopting to proactive methods of risk management rather than the later study of the issue and searching for solutions, so early actions can be taken. For Example: A person applies for a credit card in a bank, the bank can analyse previous loan history, buy now pay later scheme to completely analyse a pattern and the bank can thus identify a risk of a person being very needy for applying for a loan but a bit dicey when it comes to loan re-payment. This signifies potential risks and thus signal the institution of the risks involved in lending the customer a credit card.

3) Reduces Financial Loss:- Behavioural analysis can help an organisation to detect few loopholes in the consumers associated with risk classification and can help organisations from a very possible losing situation.

According to Kavita Dhanushkodi, Akila Bala, Nithin Kodipyaka (2025). RFM methodology is considered as the most important framework used for customer segmentation, that is grouping customers with respect to Recency, Frequency, Monetary Value which groups customers like

1) Most recent visit to the website, shop, outlet.
2) Number of times the person visits in a certain timespan.
3) Amount that a customer spends over a fixed time slot.

More such techniques include tools like Data Mining, Data Warehousing etc which collects information from different sources called as data mining which is later collected in a data warehouse which is used to collect, store and manage large amounts of information in them.

Another such study by Dr. N Kala studies a similar User Behavioural Analytics(UBA) which means to analyse various patterns in order to study abnormalities, which means unusual behaviour in order to fulfil our goal which is to detect risks, the study by Dr Kala revolves around two things

i) What does the “normal” behaviour specify with respect to each organisation .
ii) To counter or clarify the deviations from normal baseline, also explained that the not normal behaviour sometimes may even not signal a problem.

Some of the examples of UBA alerts mentioned in the study are:-

i) Alerts when user try to access the data which has rarely been accessed before that to in an abnormal time period like suppose 4AM on Sunday morning.
ii) A basic example of failed login attempts that to on an unusual time that is the time outside the normal behaviour.
iii) Alerts should be made when the user tries to copy the files from one home directory to an USB device.

2.3) Insider Threats:

For an organisation with the increasing security, risks from outsiders are very limited as most organisations security policy revolves around mitigating risks from outsiders, But, Lately risks from insiders are considered to be more critical in the IT Industry with increasing data breaches and as a result compromise the CIA Traid , also as per (Olajide O. Ogunbodede1 , Olumide S. Adewale2 , Boniface K. Alese3 , Oluyomi K. Akinyokun4, 2024) insider threats prove to be much more challenging than the outsider threats and quite logically so as insiders are often familiar with the security architecture and moreover with the strength's and weaknesses of the architecture also threat's from trusted employees compromise or normal to malicious activities which prove to be very difficult to identify and counter, so setting the parameters as normal or malicious is a very significant task to detect breaches and mitigate the risk.

Figure 2: Breaches

2021-2026

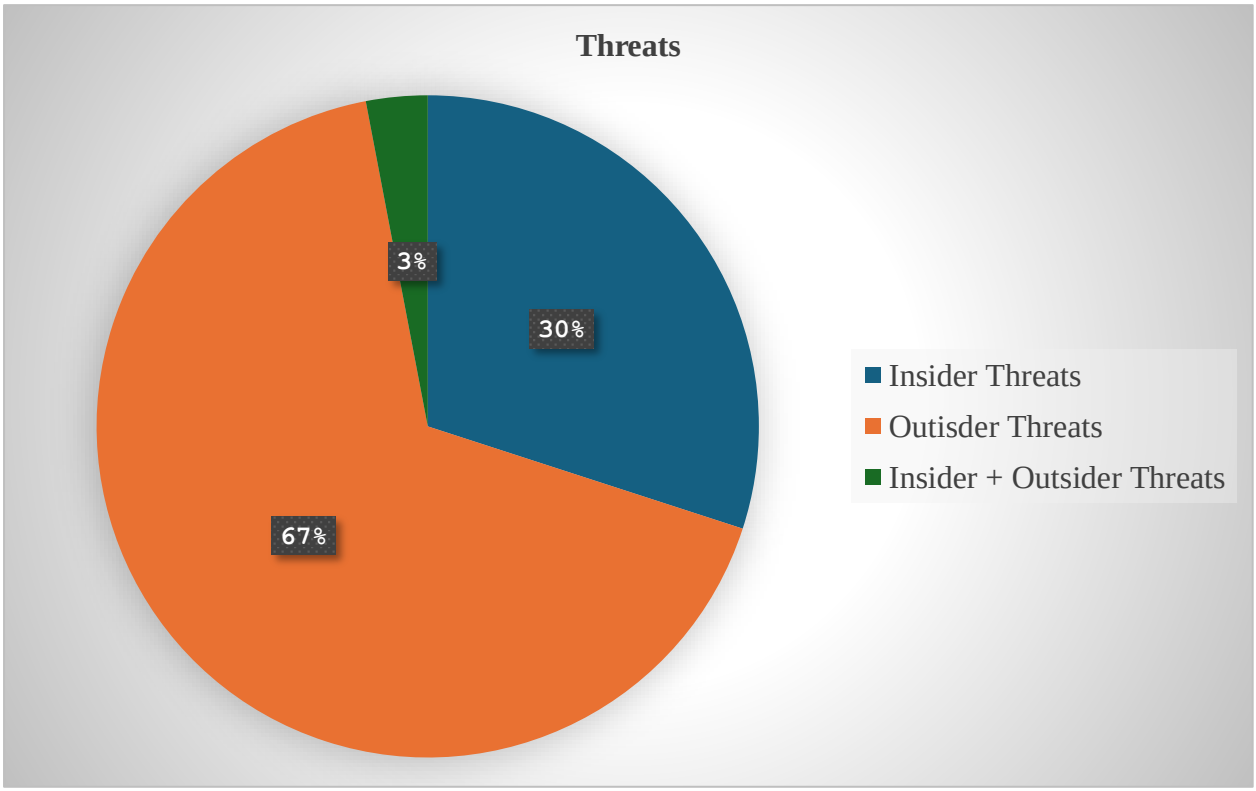
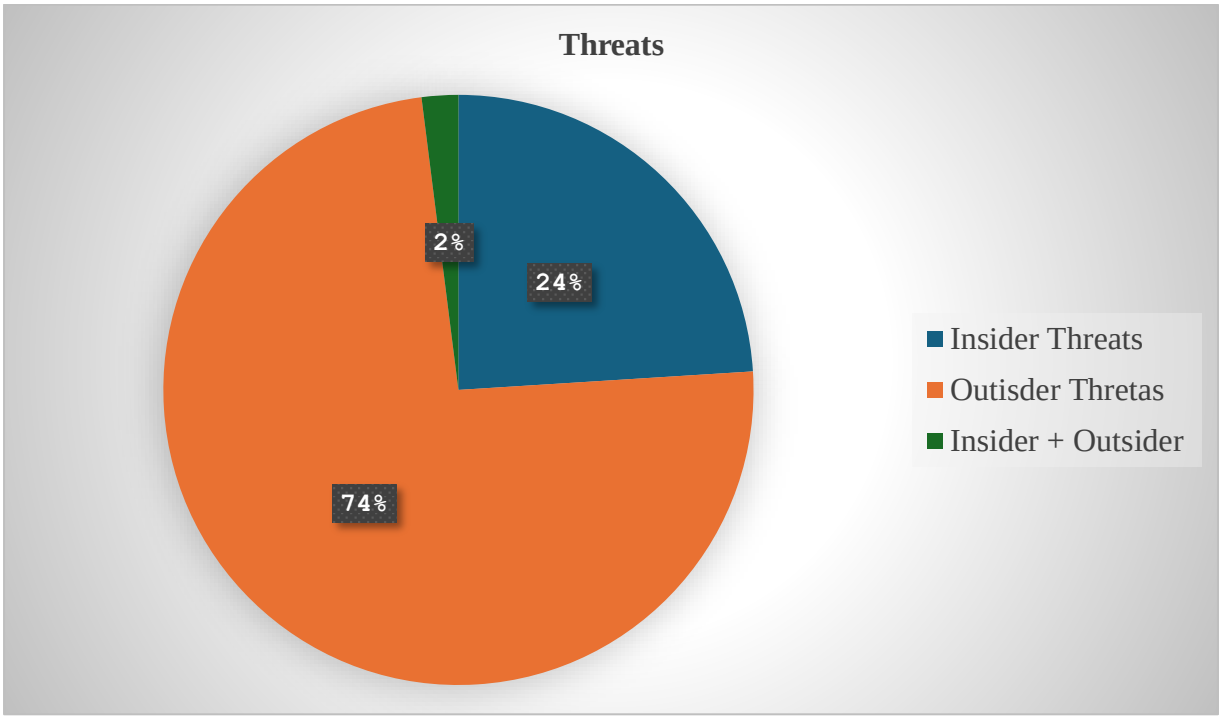


Figure 3:Breaches

2016-2021

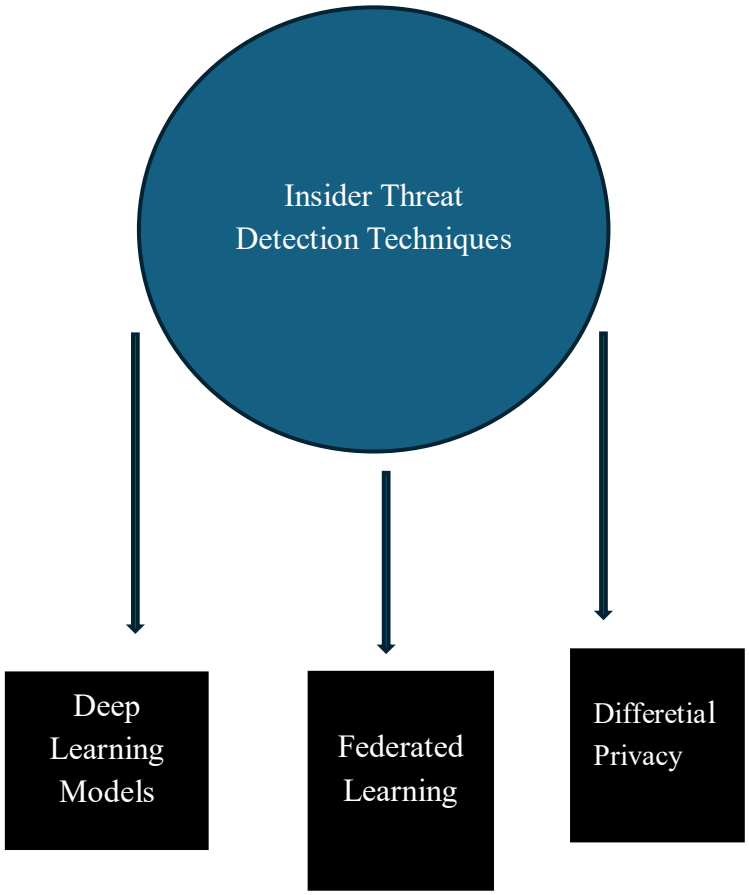


With the following graphs it becomes very convenient to say that insider threats have increased in the past 10 years with the percent increase of 7% in the insider threats in the past 5 years than the previous 5 years, we can conclude that breaches increase along with the competitive spirit in individual's. Since the pandemic individual's have lost jobs due to migration from primary occupations to new domains by individual's from the rural areas also many corporate jobs gone with the rise in AI in the same time period. All these reasons shape individual's to perform breaches in return of money or executive posts.

2.4) Insider Threat Detection:

Threat detection itself is considered a very critical job for an organisation's well-being as it is one of the domain aspects when we talk regarding the business perspective, outsider threats are very common and are planned to be taken care of when we try to construct or develop an organisation but insider threats that is the threats from the company's own employees, stakeholder's, business partner's are considered to be insider threats which cause more damage as these are individual's who know a lot about the venture and can casue damage to a lot more extent.

Figure 4: Threat Detection Techniques



As cited by Oladpo, Olabisi, Ayobami (2026) Deep learning Models, Federated Learning, Differential Privacy are the prominent techniques mentioned to overcome insider threats and its effects. The prominent techniques mentioned are Deep Learning models, Federated Learning, Differential Privacy

1) Deep Learning Techniques:- Insider threat detection earlier were classified with SIEM techniques which included comparing user activities with normalities that is everyday activities, though these weren't robust in nature and lacked complexity detection, then machine learning techniques like Bayesian Networks, Random forest techniques came into existence which proved to be great to study complex pattern but these two struggled with long term, time series data due to which neural networks were introduced like RNN and LSTM which analysed user behaviour over time and hence made it easy to compare past and future patterns with the existing state.

2) Federated Learning:- A machine learning technique and a multiple system approach which gives us result by training together a model on multiple machines without sharing raw data within themselves

so ultimately it ensures data privacy and helps improve model accuracy, there are 3 types of Federated learning approaches namely Horizontal FL, Vertical FL, Federated Transfer Learning.

3) Differential Privacy :- Technique where noisy data is added to the existence data set which is helpful as information can't be identified by model outputs and predominantly used alongside sensitive data and combined with other models which can also keep accuracy intact along with privacy.

2.5) Financial Behaviour:

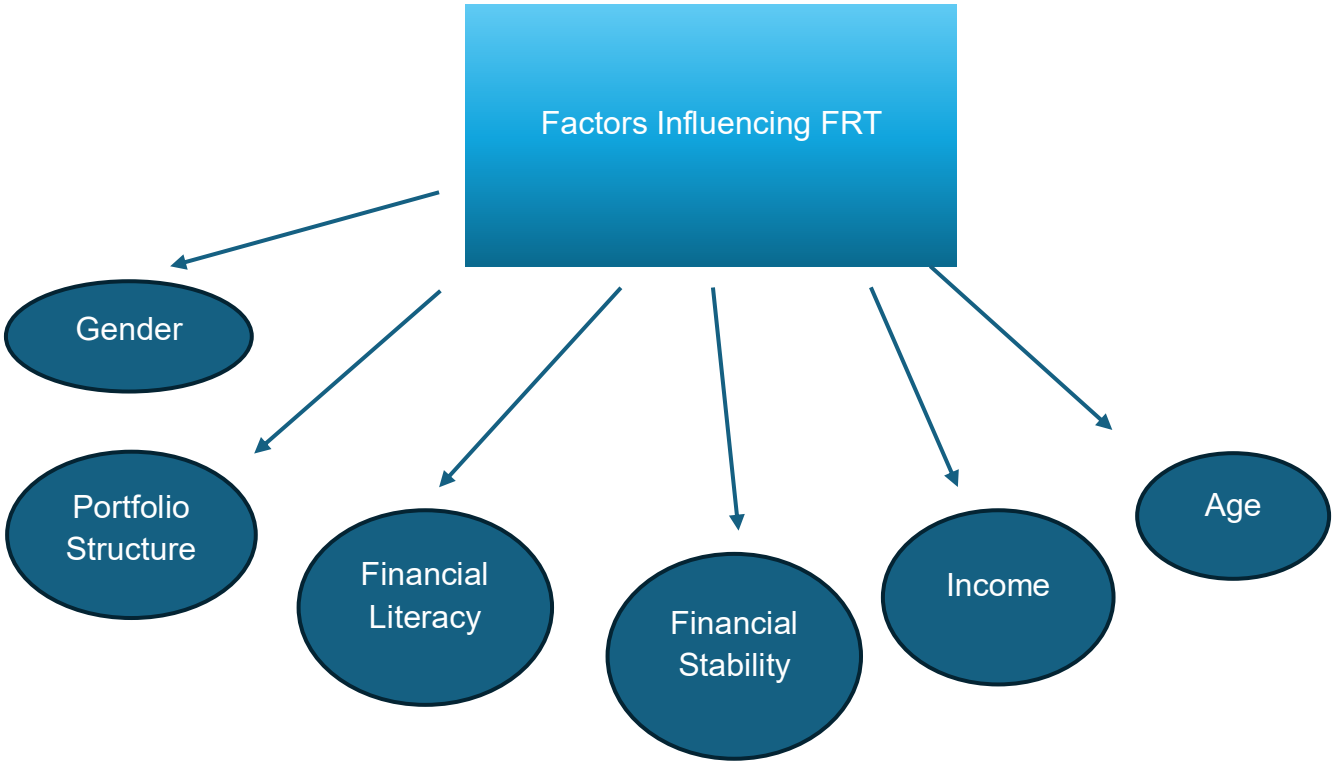
A study by Lajuni et.al (2017) claims that educators have a massive impact on financial decisions and financial stress among undergraduates in Malaysia, the study claims that the education system is majorly responsible for bankruptcy among youths where it stated that 11 youths on an average are bankrupt in Malaysia, also it stated the importance of educators in an individual's life from primary schools as children spend their maximum time in school with the educators rather than with their parents, be it in school, tuition's where their financial education has already started, so it also signifies the need of change in the education system and the added importance of teachers in the financial education of children's as well as Lajuni et. Al(2018) state's behavioural traits have a greater impact on personal financial behaviour of individuals as compared to financial knowledge on Malaysian millennials.

Financial Behaviour is the combined effect of financial knowledge and the person's self-confidence while spending money (Ramalho, Forte, 2019) . Where Tang and Baker(2016) believes that psychological traits, personal self-esteem have a more impact on financial behaviour's A representative dataset of united states, said that adults self-esteem is more influential on financial behaviour directly as well as indirectly with respect to subjective financial knowledge

2.6) Financial Risk Tolerance :

The word itself tells us the ability of an individual to deal with financial losses, risks from an investment point of view, financial risk tolerance is also considered to be an important variable to define a person's financial decisions (Grable,2016). A south- African study states that financial risk tolerance is usually measured by short term and long term investments in order to profile the customers by financial institutions, Mankuroane (2021)

Figure 5: Factors Influencing FRT:



Irandoust (2017) mentioned that gender, portfolio structure, Financial Literacy, Income ,Age are one of the most common factors that affect Financial Risk Tolerance (FRT) . Furthermore, it is mentioned that risk aversion the opposite of risk tolerance proves to be handy to explain life chores with respect to economics and daily life decisions. Generally risk revolves around three things Risk Tolerance, Risk Preference, Risk Aversion (Ahmed,Warokka,Lestari, 2020), Also, it is said that higher the level of risk tolerance the investment decisions would revolve around higher risk investments, for example, real assets. The lower the risk tolerance of a person the safer option of investing would be adopted like, for example, banking services. Nobre et.al(2018) believes that risk tolerance can be classified into a model called riskPACK which means:-

1) Risk propensity:- The correlation between risk and return a person is capable of accepting, as for higher returns usually higher risk is involved.
2) Risk attitude:- Refers to the mental and emotional capabilities of a person to work with risk.
3) Risk capacity:- The current financial stability of an investor to invest in risks, current income, risk mindset, regular flow of money are important factors involved in risk capacity.
4) Risk Knowledge:- The most important factor accompanying risk, the understanding of risk and returns.

2.7) **Gender Inequality towards behavioural finance:**

The financial decisions tells a lot a person's background, environment an individual lives in, every family or every organisation has set certain norms of financial spendings which each member of the family or organisation has to stick with, generally spendings differ from person to person with respect to background, earnings etc. A very generic form of discrimination is gender where one category predominantly spend similarly with respect to their financial stability. In the banking sector, there is a difference in terms of spending between both men and women due to lesser income, lesser education. Yiu et. al(2007), Polasik and Wisniewski (2009) state that the banking sector is more dependent on men, where instances have not solely been given from their preference of banking mode, but also from their hands-on in using computer's, women if earning are not so frequent in depositing money in financial resources, as they are considered secondary income earners and spend most of their money in buying food and grocery items required in the house rather than investing and expecting returns (Johnson,2004). In this idea [Christensen, 1992; O'Quinn & Faber, 1989] has stated that women show relatively higher compulsive buying behaviour than most men. Also, if we study ancient literatures it has been said there is no clear difference between how men and women buy unplanned things when the sample size of the number of purchases were evenly sorted. Gender inequality in spending and financial decisions are not only because of income inequality between both the respective genders but also due to various other psychological, socio-economic factors that affect both the classes uniquely [Faber, 1992]. More importantly in a country like India, the concluded perspective of (Tahira, Olive 2000) is very relatable that financial stress among women is very much likely to affect their work performance, but still less likely to affect her relationship between her family and loved ones.

2.8) Research Gaps:

Existing studies have already worked on all the related concepts like Financial Risk Profiling, Machine Learning for behavioural finance, NLP for behavioural finance, where extensive researches have been developed from previous market datasets. They have primarily focused on datasets that only revolve around general importance of behavioural finance, techniques used to classify risk, major studies don't rely on multiple behavioural factors, where we will work on **Risk Appetite Score, Stability Index, Diversification Index, and Monitoring Intensity Score** using clustering techniques in order to simplify risks.

3) **Literature Review Table:**

SR NO	TITLE	SUMMARY
1	Kattoura, Salem. *Investor risk profiling: an enhanced behavioral finance perspective and technique for identifying client risk profiles*. University of Wales Trinity Saint David (United Kingdom), 2022.	The study is associated with risk fundamentals, profiling risk majorly in three terms risk tolerance, risk appetite, risk capacity i.e. risk you are willing to take, risk you can afford to take, risk you need to take.
2	Ishii, Shogo, et al. *Capital account liberalization and financial sector stability*. Vol. 211. Washington, DC: International Monetary Fund, 200	Talks about capital liberalization, which looks forward to the topic of uneven flow of information and money due to uneven development in a particular country
3	Kimball, Miles S., Claudia R. Sahm, and Matthew D. Shapiro. "Imputing risk tolerance from survey responses." *Journal of the American statistical Association* 103.483 (2008): 1028-1038.	The authors say that different individuals have different psychological and behavioral traits to risks involved, hence grouping individual's who react similarly to risks involved becomes a crucial part of risk profiling.
4	Nayak, Saugat. "Developing predictive models for financial stability: Integrating behavioral analytics into credit risk management." *Journal of Artificial Intelligence & Cloud Computing* 3.5 (2024): 2-10.	Traditional financial models fails to identify individual's correctly lacking the behavioral aspect in the study, where consumers dynamic reaction to risks becomes a very important part when creating a model integrating behavioral analytics into traditional risk management study.
5	Varma, Santhosh Chitraju Gopal. "Artificial Intelligence for Behavioral	Behavioral risk is a non-quantifiable and unavoidable measure, basic human lapses during investing ism termed as behavioral risk.

	Risk Analysis in Finance." (2025).	
6	Qadri, Hussain Mohi-ud-Din, and M. Ishaq Bhatti. "Islamic Finance in the Modern Era." *Retrieved January* (2025).	Talks about how modern day innovative financial products affect the religion driven Islamic financial industry.
7	Suresh, M., et al. "Analyse customer behaviour and sentiment using natural language processing (NLP) techniques to improve customer service and personalize banking experiences." *Educational Administration: Theory And Practice* 30.5 (2024): 8802-8813.	Drives the importance of natural language processing in the financial industry field, regarding how various customer feedback techniques fails in understanding the exact motto or the emotions behind text and NLP comes to play.
8	Dhanushkodi, Kavitha, et al. "Customer behavior analysis and predictive modeling in supermarket retail: A comprehensive data mining approach."	Customer segmentation traditionally revolved around RFM analysis, creating a loophole in behavioral analysis which is overccomed by predictive modeling.
9 10	Kala, N. "COMBATING INSIDER THREAT USING'USER BEHAVIOURAL ANALYTICS'." Ogunbodede, Olajide Olawale, et al. "Insider Threat Research: A Review of Deep Learning Approach."	User behavioral analytics used to study the abnormalities in investing, also suggesting that normal is not the same for everyone, hence also having different abnormalities. Insider threats prove to be much more hazardous to an organization, as compared to threats from the outside, keeping in my mind the familiarity to the environment.

11	Ogunbodede, O., et al. "Insider threat detection techniques: Review of user behavior analytics approach." *International Journal of Research in Engineering and Science* 12.9 (2024): 109-117.	Discusses the various techniques that can over come insider threats to an extent, majorly deep learning techniques, federated learning and differential privacy .
12	Adeduro, Oladapo, Olabisi Josh-Falade, and Ayobami Mesioye. "Proactive Insider Threat Detection Framework: An Explainable AI and Behavioral Analytics-Driven Approach." *Journal of Future Artificial Intelligence and Technologies* 2.4	Various insider threat detection policies with the help of AI and Behavioral analytics are discussed.
13	Lajuni, Nelson, et al. "The role of educators and the influence of financial behavior on personal financial distress among undergrad students of public universities in Sabah, Malaysia."	Studies the strong influence of educators, teachers among financial decisions taken by youth.
14	Lajuni, Nelson, et al. "Religiosity, financial knowledge, and financial behavior influence on personal financial distress among millennial generation."	Defines that a person's behavioral traits impact more on persons financial decisions than the financial knowledge an individual has.

15	Ramalho, Thiago Borges, and Denis Forte. "Financial literacy in Brazil–do knowledge and self-confidence relate with behavior?."	It displays a positive influence or relationship between all the three: Knowledge, Self Confidence, Financial Behavior.
16	Tang, Ning, and Andrew Baker. "Self-esteem, financial knowledge and financial behavior." *Journal of economic psychology* 54 (2016): 164-176.	It says self-esteem plays a negative role on the financial behavior of a person indicating inverse decisions, despite strong financial knowledge.
17	Grable, John E. "Financial risk tolerance." *Handbook of consumer finance research*. Cham: Springer International Publishing, 2016. 19-31.	The research tells us about financial risk tolerance as a crucial component of financial knowledge, provides models in order to enhance financial risk tolerance.
18	Mankuroane, Tlotlego Evadia. *Analysing the factors that influence investment intentions in South Africa*. Diss. North-West University (South Africa), 2021.	Discusses risk tolerance, personality differences, and individual bias in order to profile or group individual's by their financial wellbeing.
19	Irandoust, Manuchehr. "Factors Associated with Financial Risk Tolerance Based on Proportional Odds Model: Evidence from Sweden." *Journal of Financial Counseling and Planning* 28.1 (2017)	Studies the factors influencing risk tolerance, also talks about the inverse that is known as risk aversion, Factors are said to be demographic in nature like gender, income and age etc.
20	Ahmad, Gatot Nazir, Ari Warokka, and Irna Puji Lestari. "Financial risk	Risk Tolerance: The amount of risk an individual can afford to take, Risk Aversion: The exact opposite of risk

	tolerance analysis of Indonesian retail investors." *Humanities & Social Sciences Reviews* 8.4	tolerance, Risk Preference: The type of risk a person feels comfortable while handling is risk preference.
21	Grable, John E., Abed Rabbani, and Liana Holanda Nepomuceno Nobre. "Comfort with risk: a new perspective on risk tolerance." *Research Handbook on Personal Finance* (2026): 245-264.	Explains certain characteristics of risk tolerance: riskPACK Risk Propensity. Risk Attitude. Risk Capacity. Risk Knowledge.
22	Yiu, Chi Shing, Kevin Grant, and David Edgar. "Factors affecting the adoption of Internet Banking in Hong Kong implications for the banking sector *International journal of information management*	Clarifies the dominance of men in the financial industry, with studies states women as secondary income earners which strive for basic needs rather than showing dominance in the investment sector.
23	Polasik, Michal, and Tomasz Piotr Wisniewski. "Empirical analysis of internet banking adoption in Poland." *International Journal of bank marketing* 27.1 (2009)	Poland as a country shown total dominance of men in the banking industry, where internet banking users are mostly men majorly also due to several other factors like education, work and home balance etc.
24	Johnson, R. Burke, and Anthony J. Onwuegbuzie. "Mixed methods research: A research paradigm whose time has come." *Educational researcher* 33.7 (2004)	Shows differences between women spending bias and men spending bias, which similarly says that women work for basic needs rather than growing through the investment banking sector.
25	Faber, Ronald J., and Thomas C. O'Guinn.	Dominance of women in compulsive buying behavior is seen in the research, also stating that there is no

	"Classifying compulsive consumers: Advances in the development of a diagnostic tool." *Advances in consumer research* 16.1 (1989).	specific reason for the finding but only general rule of nature and how humans are made.
26	Faber, Ronald J. "Money changes everything: Compulsive buying from a biopsychosocial perspective." *American Behavioral Scientist* (1992):	This shows the slight incline of women in compulsive buying behavior and considers socio-economic factors the most dominant of the factor influential in women.

4) **Methodology:**

4.1) Data Collection: -

The dataset however is obtained from Kaggle, which focuses on investors or banking customers, who have been generally examined to understand financial behavior trends as such, the dataset moreover focuses on a lot of attributes precisely 24 of them with 12001 records, that is 24 columns and 12001 rows and hence the dataset could turn out to be of great help in order to examine all the investors closely and carry out the project objectives. The entire data required for our analysis consisted of a dataset named " **Finance Trends.csv"** which mainly focused on columns like gender, age, investment avenues, mutual funds, equity market, debentures, government bonds, fixed deposits, PPF, gold, stock market, factor, objective, purpose, duration, invest monitor, expect, avenue, objective's, reason equity, reason mutual, reason bonds, reason fixed deposits, source being the primary columns present in the dataset.

4.2) Data Preprocessing and Cleaning: -

Finance_Trends.csv our project name with the csv extension was imported using the pandas function (pd.read_csv,),to help manipulation with the data inside the csv file, the first part of preprocessing or cleaning was to delete the missing values from the dataset, another thing was conversion of the needy columns into numerical where it was required.

4.3) Feature Engineering: -

Capturing investor behavioral traits was the entire motto of the project which was more accurately carried out using feature engineering, feature engineering is the process of converting raw data into features which prove to be meaningful in calculating the desired outcome. The engineering process revolved around creating 4 features which were categorical as well as numerical, also all the 4 were created in order to help the machine learning process directly without the competitiveness of manipulating the existing columns directly at the time of calculating the clustering results.

\

Steps involved in feature engineering process: -

1) Selecting useful variables out of the lot, which is converted into features for the machine learning process
2) The features created are later used as inputs in the test-train-split method to train the models using our features.

4 features were created namely Risk Appetite, Stability Index, Diversification Index, Monitoring Frequency

Risk Appetite: It's the quantified risk calculated based on the investor's choice of investments, it's the measure of how confident or how willing is a person to take financial risk. Created using equity + mutual funds

Equity: Usually, higher risk can also reward in higher returns.

Mutual Funds: Usually moderate to low risk involved

So, the combination of both can usually signal and tell us the risk-taking capacity of the investor.

Stability Index: It was basically an index created for safe investments as people who invested in more stable and safe investments like FD's and PPF's were generally mentioned to have high stability score.

Fixed deposit's: - Generally, fixed deposits are considered one of the safest investment strategies where you can guarantee stable returns over time and is considered fairly low risk as compared to other investment options.

PPF's: - It stands for public provident funds this is mostly government based like a government scheme made for safe and stable investments, supports long term investments and offers capital security with guaranteed results.
Investors with high Stability Index are considered pretty much safety focused rather than being return focused.

Diversification Index: This generally was created to measure the portfolios, and can signal the multidimensionality of respective portfolio's by capturing various high risk and low risk feathers in an individual's cap. In short, Diversification Index means the diverse investing options a person or an investor is interested in, like more variety in investing pattern may signal more diversification index for example a person invests in mutual funds as well as equity as well as PPF.

Practical Example:

Investor	Equity	Mutual funds	PPF	FD	Diversification Index
1	Yes	Yes	No	No	2
2	No	Yes	Yes	Yes	3

Table 3: Investor Profiling

Monitoring Frequency: In nature, this is one of the most misleading variable which doesn't affect our numbers directly but has a very big passive impact on investor personality traits. Monitoring Frequency means how often does an investor checks his or her investments as generally,

High monitoring score: Signals symptoms of an active investor who is more likely to get good returns rather than low or moderate investors.

Similarly moderate as well as low monitoring score investors are considered to be less active about their investments and might miss the opportunity to yield high returns being less active about their investments.

After feature engineering one more important step initiated was the MIN-MAX scaling that is we converted all the values in the dataset to a scale of 0 to 1, this was necessary to ensure unbiased clustering as larger numbers usually make the clusters drift away from the exact project plot and no major cluster dominates the range of clusters.

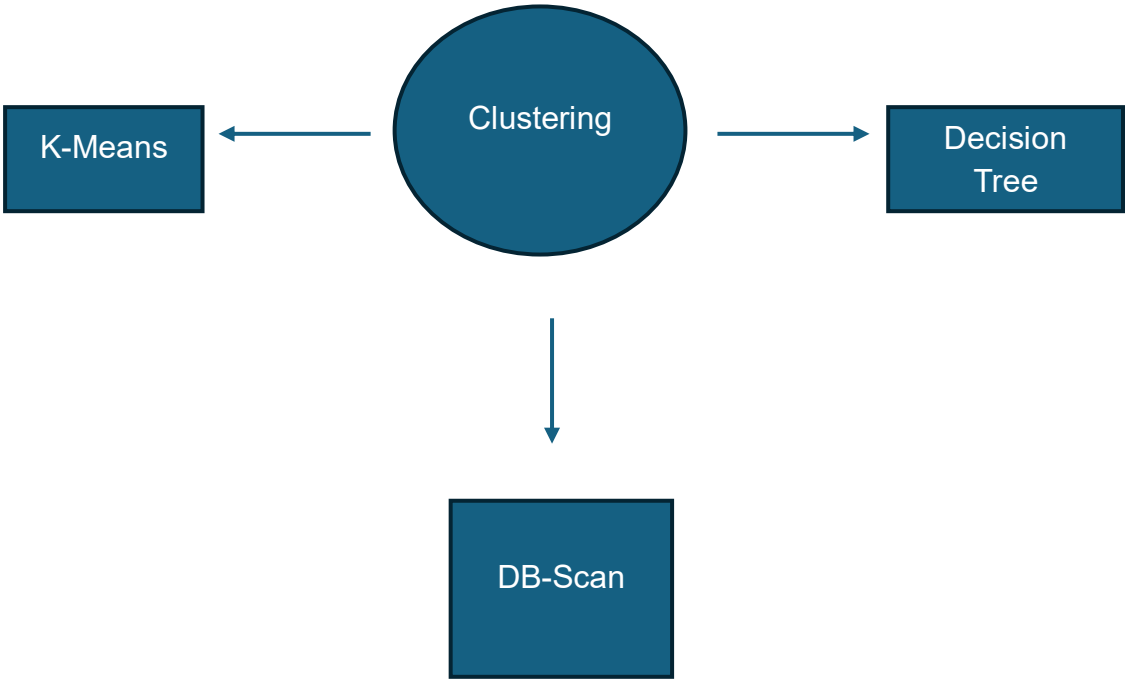
All the four engineered variables together justify all the needs of analyzing risks where we examine risk taking, diverse behavior while investing, conservative behavior using stability index, monitoring tendency with the help of monitoring frequency.

4.4) Clustering and results: -

The project focuses entirely on grouping the investors in various categories which predominantly show similar investing patterns, for this task at first an unlabeled dataset was present which means we did not have any predefined labels to train our models, so unsupervised machine learning technique like clustering were used to train our model and reach our end result.

The various clustering techniques used in our project were:

Figure 6: Clustering & Decision Tree



I) K-Means clustering algorithm works on the principle of dividing the dataset present into a fixed number of clusters, practically, clusters are groupings of data that nearly fall under the same range and can be grouped together as whole,

A very basic example could be the **grading system** used in schools like 80-90 you fall under the A2 category so basically, students are grouped in consideration to their scores in a particular subject, so as mentioned the K-Means algorithm groups clusters like

. number of clusters= 3

. matching identical or nearly identical scores together

. Matching clusters until we have a near stable cluster count at all the clusters formed

In our project we have used 3 clusters, the number of clusters were decided based upon the general behavioral and also basically keeping in mind their respective financial condition as well, the 3 clusters finalized were named cluster 0, cluster 1, cluster 2 which categorized the investors into various risk profiles

II) DB-Scan clustering was the second clustering algorithm used in our project, the success of any algorithm can be judged only when it's compared to different algorithms of the similar kind so running both the clustering algorithms and validating which algorithm proves handy to reach our end goal was the primary objective, unlike K-Means algorithm the DB-Scan algorithm

. Doesn't work on any fixed number of clusters

. It groups the clusters or investors in our case, in terms of density or the spread of data

4.5) Limitation:

DB-Scan on definition itself gives us an idea that it doesn't perform so well on densely spread data which consists of outliers as compared to K-Means because if we have 2/3 sample points near a cluster the K-Means algorithm will fetch that data to the nearby clusters but as DB-Scan is a density based algorithm this ignores the data point's as outliers and only focuses on points which are densely populated at a particular spot, so in action to this behavior the DB-Scan algorithm wasn't able to form meaning-full clusters and also if clusters were formed they were pretty biased due to which we can term DB-Scan as a failure as compared to the K-Means algorithm.

III) The decision tree algorithm here does the work of validating and explaining why a particular investor falls in a cluster what are the behavioral traits that separate the investors in the three clusters, so as we already have the K-Means algorithm results and have proved it to be the most effective clustering algorithm for our project, the decision tree works on the K-Means algorithm results, so basically we grouped particular investors in three cluster that is cluster 0,1,2 respectively now the decision tree expands further that what are the behavioral differences that make the investors fall under one particular cluster. The decision tree algorithm works on the principal of x and y variables which means input features(independent variable) and output features(dependent variable) respectively, where, X=scaled features and Y=K-Means cluster where later we used the train test split to train the model with the decision tree classifier where later after the training and testing results we got the decision tree score as 1.0, which normally shows overfitting but here in our project as we already have the results and the decision tree is only used for validation, we can conclude that the clusters that are formed by the K-Means algorithm are clearly understood and are easily separable. Which tells few important things about our project that the features created aren't random, also the K-Means algorithm is doing the job perfectly.

IV) Silhouette Score was the technique used to validate the clustering algorithms used, silhouette score basically tells us whether the clustering done matches the expected results or not, it checks the distance between the original cluster and the investor and the distance between the neighboring cluster and one particular investor, like for example if a person A falls under the cluster 0, the clustering can be said valid if A is closer to its own cluster than any other cluster that has been formed, since DB scan didn't form any meaningful clusters we weren't able to compute any silhouette score for the DB-Scan algorithm.

V) Concluding, the methodology focused on deriving features that were useful for model execution and giving effective results, clustering algorithms were used instead of basic Machine Learning algorithms due to the vast diverse and large dataset that was present, K-Means, DB-Score and later decision tree for cluster validation was used which were built upon the 4 new variables that were created from the existing dataset.

5) Result And Analysis:

5.1) Cluster Summary and Risk Mapping:

```
cluster_summary = df.groupby('DBSCAN_CLUSTER')[[
    'Risk_Apetite',
    'Stability_Index',
    'Diversification_Index',
    'Monitoring_Frequency'
]].mean()

print(cluster_summary)
```

```
                Risk_Apetite  Stability_Index  Diversification_Index  \
DBSCAN_CLUSTER
0                   2.996792         2.789958               2.190167

                Monitoring_Frequency
DBSCAN_CLUSTER
0                            2.00275
```

Figure 7: Clustery Summary (DB-Scan)

```
cluster_summary = df.groupby('KMeans_Cluster')[[
    'Risk_Apetite',
    'Stability_Index',
    'Diversification_Index',
    'Monitoring_Frequency'
]].mean()

print(cluster_summary)
```

```
                 Risk_Apetite  Stability_Index  Diversification_Index  \
KMeans_Cluster
0                    2.733295         2.635088               2.000000
1                    2.731034         2.630126               2.000000
2                    4.586006         3.738484               3.330612

                 Monitoring_Frequency
KMeans_Cluster
0                            3.000000
1                            1.499854
2                            1.994752
```

Figure 8: Cluster Summary (K-Means)

Based on the results we get a clear indication why K-Means really stands out from DB Scan cluster, we can identify that K-Means has segregated the entire dataset into three clusters with the scores mentioned under the cluster names, which signals that K-Means was successful in identifying the differences in behaviours within the dataset, whereas its counterpart DB-Scan wasn't able to identify any cluster which eventually meant that all the investors typically behaved in a similar manner.

When we compare both the results with respect to the main aim of our project we intend to separate various investors with respect to the risk associated with their behaviour that is we want to profile the investors with respect to risk but as DB-Scan didn't deliver any differences within their attitude we can declare K-Means as the eventual winner.

```
#DBSCAN SCORE(ONY IF MORE THAN ONE CLUSTER EXISTS)
labels = df['DBSCAN_CLUSTER']
unique_clusters = set(labels) - {-1}

if len(unique_clusters) > 1:
    score = silhouette_score(scaled_features, labels)
    print("\nDBSCAN Silhouette Score:", score)
else:
    print("\nDBSCAN did not form meaningful clusters")
```

```
DBSCAN did not form meaningful clusters
```

Figure 9: Silhouette Score (DB-Scan)

```
#evaluation (silhouette score)
kmeans_score=silhouette_score(scaled_features,df['KMeans_Cluster'])
print("K-Means Silhouette Score:",kmeans_score)
```

```
K-Means Silhouette Score: 0.46946648623876763
```

Figure 10: Silhouette Score (K-Means)

For clustering Silhouette Score is the metric generally used which tells us about the quality of clustering which took place, theoretically it ranges from -1 to 1 where a score close to 1 shows perfect clustering whereas negative score justifies that the clustering wasn't appropriate, From the visuals above we can clearly get an idea that DB-Scan wasn't able to segregate the users properly and hence no separation was being seen, whereas on the other hand K-Means got a very acceptable silhouette score close to 0.47 which indicate decent clustering where we were able to cluster or group the investors with respect to their behavioural traits, although seeing a bit of overlap which in spite shows that the K-Means behaves perfectly for a topic like ours in spite of being a drawback.

The K-Means algorithm was the most successful algorithm, as DB-Scan algorithm didn't produce meaningful clusters, we had a total of 3 clusters based on the similarly behaving data points we passed to the algorithm, later there was a need to further specify the investors as high risk, low risk, moderate risk based on

the variables that was created, this entire process is termed as risk category mapping where, we move from the clusters created from K-Means to meaningful risk categories which signifies the core of our project, so adding the data frame risk score was important as it consisted of the normalized values of the variables that were created, the risk scores were created with the formula

```
normalized['Risk_Score'] = (
normalized['Risk_Apetite'] * 0.4
normalized['Stability_Index'] * 0.3
normalized['Diversification_Index'] * 0.2
normalized["Monitoring_Frequency"]* 0.2)
```

1) The Risk Appetite gets the highest weight as it is the strongest behavioral indictor and plays a vital part in showing a person's tendency to directly take risks and increases the risk score directly, it showcases the person's tendency to take direct risks and portrays what aggressive investing pattern really is.

2) Stability Index the second investment variable gradually reduces the risk score because stability index ensures conservative investing and normally seen as an asset to reduce risk levels, it contains safe investing options like Fixed deposits, PPF's which increases the stability in investing, that is it lowers the risk of an investor suffering loss.

3) Diversification Index being the third variable created didn't show massive impact on risk score, but still proved to be an important indirect factor which showcased a person's diverse behavior while investing which significantly reduced the chances of the investor getting bankrupt and can stay steady due to the spread investment choice that were made, so Diversification Index like Stability Index also proved to be an instance that reduced risk, although being less dynamic.

4) Monitoring Frequency was the variable created which showcased the persons tendency to monitor or validate the investments, where high monitoring frequency ultimately suggested or gave us a clue about the conservative behavior amongst the investors, as continuous monitoring showed the tentativeness of facing losses and hence higher monitoring score will eventually result in a person opting for safe investments, so Monitoring Frequency very similar to Diversification Index didn't show direct but signaled conservative behavior and ultimately promoted safe investments.

Table 4: Significance of Weights

SR NO	Variable Name	Explanation
1	Risk Appetite Score (+4)	Directly influences Risk, so has a positive impact on the Risk Score
2	Stability Index (-3)	Direct Impact on conservative behavior, resulting in negotiating risk
3	Diversification Index (-2)	Signifies multi-investment patterns ultimately resulting in enhanced conservative behavior
4	Monitoring Score (-2)	Carefulness in investing, being conservative and frequent checks of investments promoting safe investments and reduced risks.

Later, 3 various clusters were created using the normalized score using the existing features i.e the clusters created using the K-Means algorithm and the risk score later created, this forecasted the need for cluster mapping that is to map the risk score with various entries and form the new 3 clusters, namely high risk clusters, low risk clusters, moderate risk clusters, these clusters were created using the idxmax() and the idxmin() function which means the index of the maximum value and index of the minimum value respectively, which matches the high risk scores and the low risk scores together and groups them further as high risk clusters and low risk clusters respectively, later the moderate clusters were named creating a set of 3 of the normalized indexes and then separating the high and the low risk cluster matches from them, with this the one remaining was classified as moderate risk cluster.

The mapping process eventually gave the numbers like:

Table 5: Count of risk types

Risk cluster	Count
Low risk clusters	6828
Moderate risk clusters	3831
High risk clusters	1341

With the distribution visually seen as:

Figure 11: Distribution of risk categories

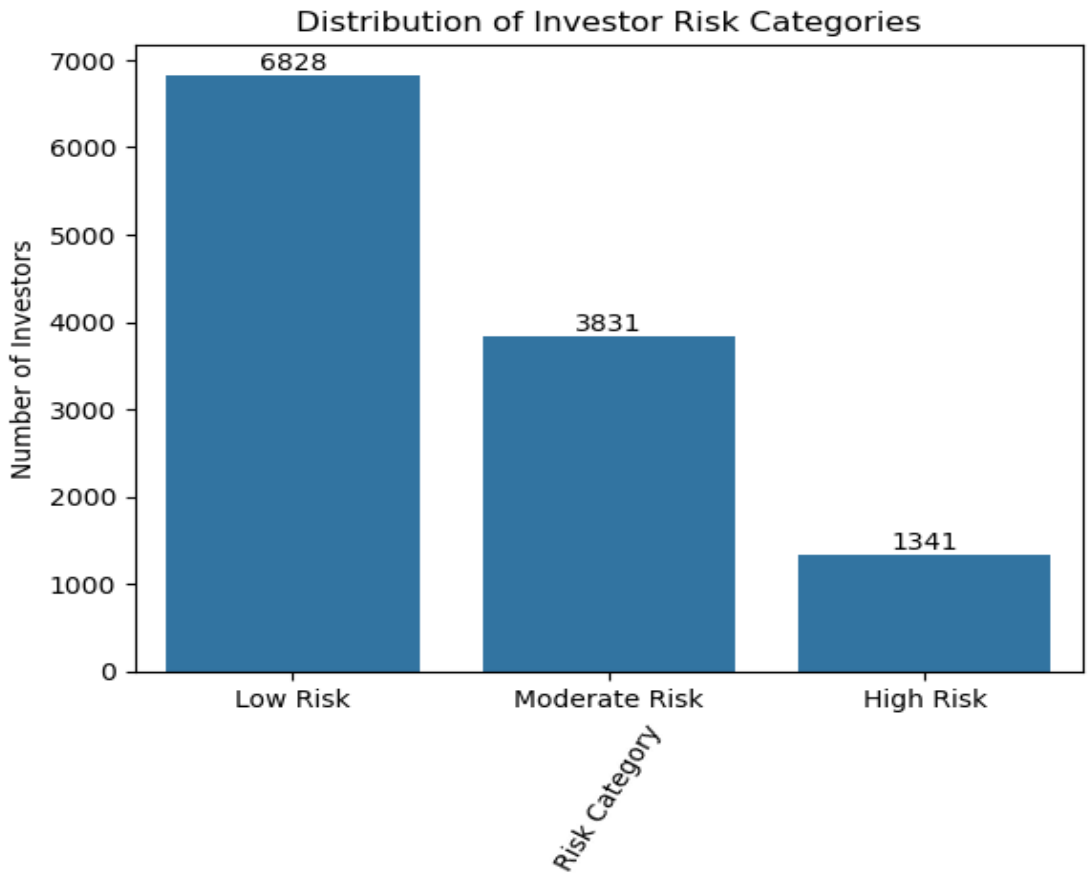


Figure 12: Motto for risk

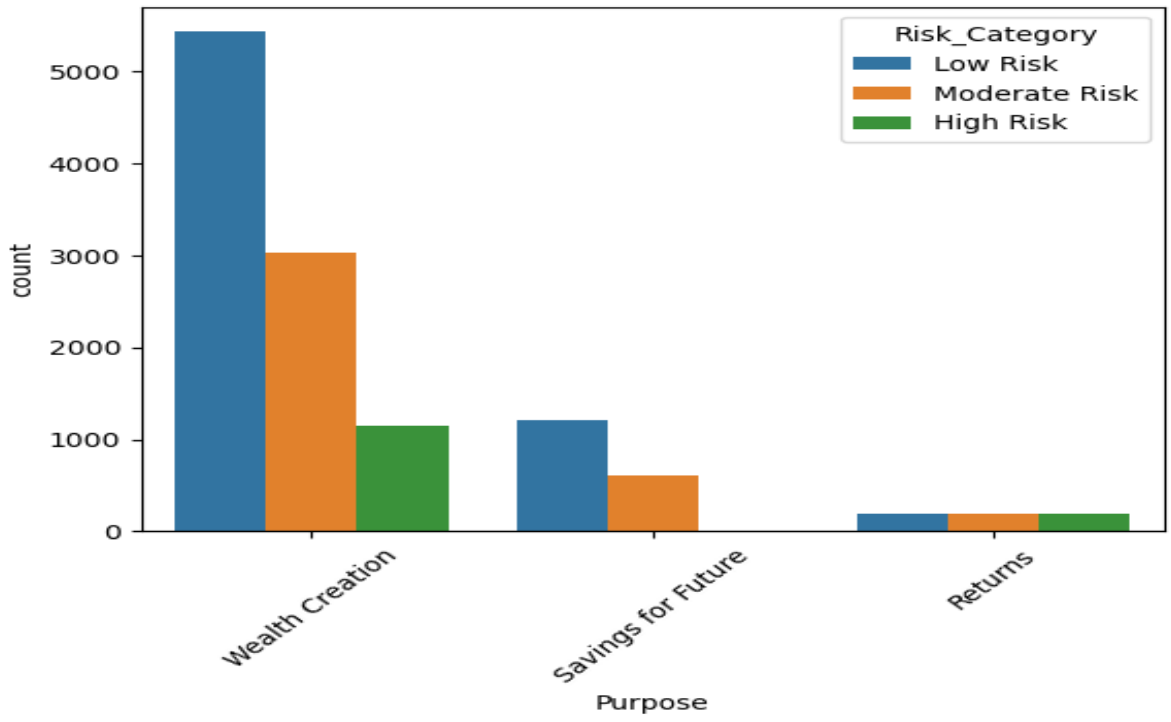
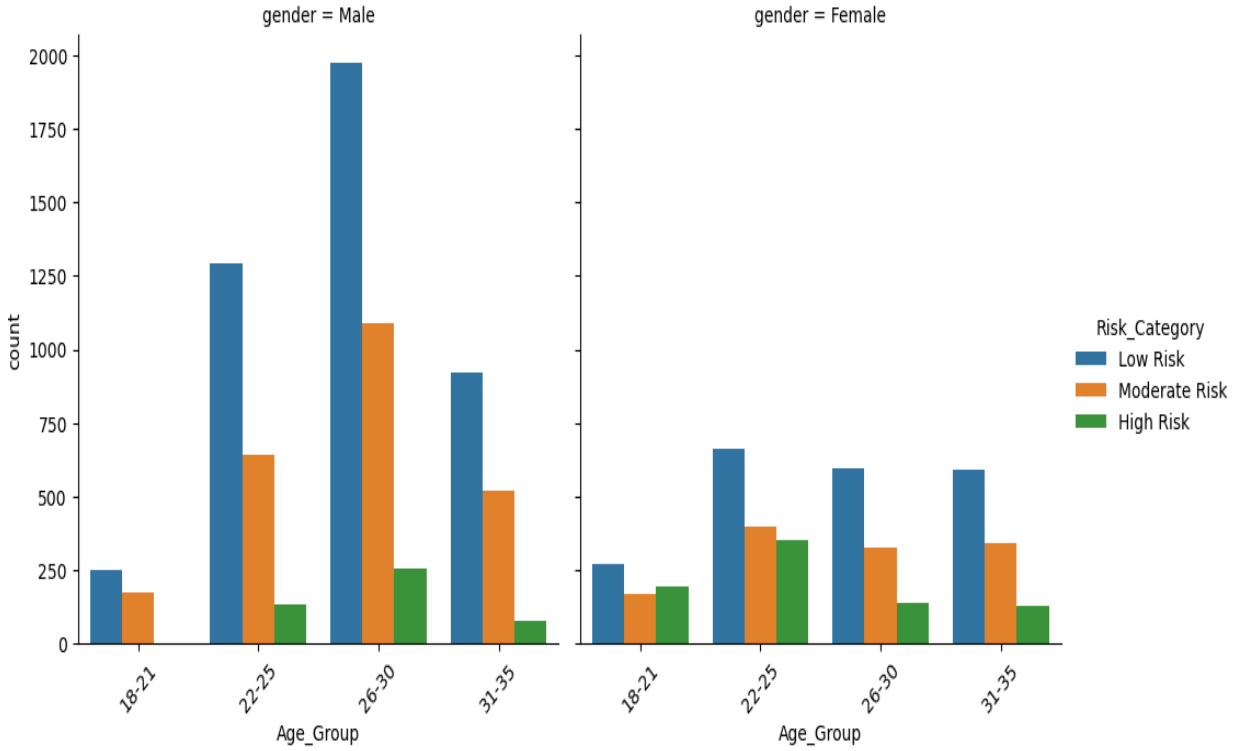


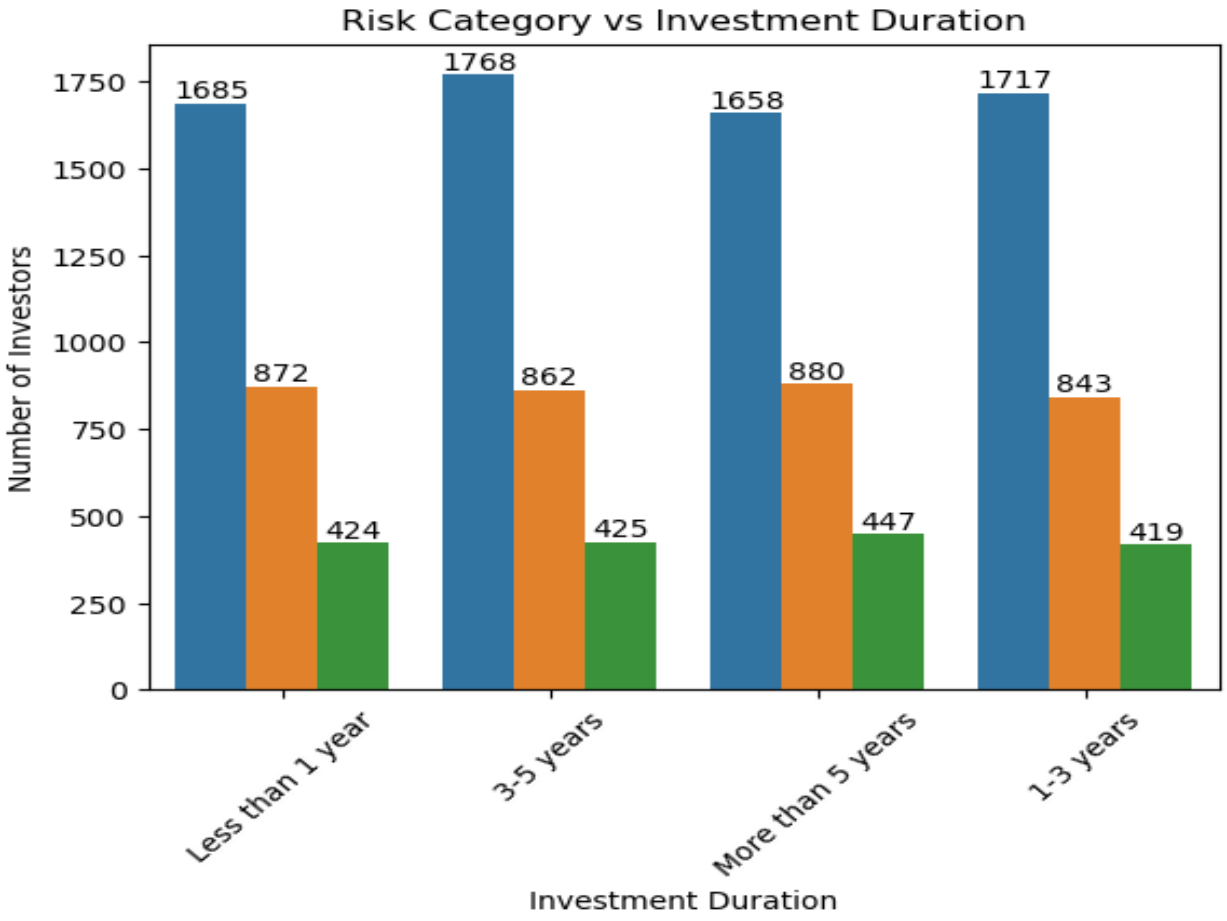
Figure 13: Gender of risk takers



The entire distribution of our dataset were split into the three broad categories and the results were quite prominent towards one side with almost 60% of the investors belonging to a category, that is the low risk criteria which also gives us an impression about the risk free nature that the investors possess toward investing. High risk investors being only 10% of the total investor's in our dataset, a slight spike has been seen in high risk investors from the age group 26-30 in men, and the group 22-25 in women, which also simplifies the working dynamics of the country like India, where women are generally seen as homemakers tend to invest more in risky stuff quite earlier than men, because of the tendency to consider women income as a secondary income source and not something a typical Indian family sees as the bread and butter income, though in the 21st century this mentality is near improving, but still majority of Indians consider female income as secondary income, so women noticeably invest or dynamically invest a lot earlier than men, one more noticeable factor which supports this theory is the early adults age group that is the 18-21 age group where we see almost no men in the high risk category whereas we see more women in the high risk category than the moderate risk category which shows that the girls in the early earning stage go for high risk investments in order to make money whereas young men have an conservative approach towards earning throughout their entire life cycle where very little spike can be seen from the age 26-30 which is the early settling stage or before marriage phase a male individual is in.

One more such important stat was about the purpose of the investments, where again we had low risk investors who outnumber the moderate and high-risk investors. 3 main purposes listed in our dataset were wealth creation, savings for future and returns where the most neglected purpose was returns, which also gave us an idea about the lack of primary investors in our dataset and presence of secondary investors that is the investors who invest a small portion of their income in order to increase their portfolio, have something with them as backup to their primary methodologies. Wealth Creation as a purpose easily outnumber every other purpose listed and rightly so as even though people have a conservative approach towards investing but always look up-to some investment options which proves to be their secondary income option, usually studies show a deep logic that people invest their money whether in mutual funds whether in Fixed Deposit's in order to save for near future but the trend these days has shifted the focus from just savings to wealth creation in a minimal risk environment, as a result we see people investing for creating wealth earning money but the scalability, the longevity is the thing that is looked upon as a no compromise factor, like suppose what this eventually means that even if people are conservative in dealing in money, they do feel the need to go a step beyond savings and expand money flow coming in their way with minimal risk, which has been a pattern been seen for long between young to mid age investors, also keeping in mind factors like income inequality plays a leading role in order to look at the consistent side of investing.

Figure 14: Risk Vs Investment:



Low risk investing proved to be the most significant factor, entity from our project which showcases the conservative tendency of individuals in our dataset, High risk investing often suffers due to the uneven nature of the market where we can say that short-term investing often suffers losses due to the urge of making more money. As mentioned before almost 60% of the investors fall under low-risk investors as a similar ratio is been seen in the purpose of investing with 60% of people investing for creating wealth, which partially can be correlated with savings for future, as people have apperception that long term investments generally make more profits as compared to investments for a shorter span. Dealing with behavioral categories turns out to be tricky, as it can be segregated with many diverse parameters like faith in investing and banking, knowledge about banking products, investing and de-investing at the right time etc.

As per our analysis from Risk Category vs Investment Duration the analysis lies very parallel to each other as people investing up to 5 years are covered and we get proportionally get the same ratio everywhere, less than a year, the range from 1-3 years, 3-5 years, more than 5 years. So, this entire thing highlights **"UNCERTAINTY".** Uncertainty which can mean lapse in money management, lapse in understanding the market dynamics which proves our theory of lapse in knowledge about banking products, this can be stated because the number of low-risk investors prove to be the most significant feature in all the 4 time spans mentioned in the graph. On a general thesis we can say that people who invest for a short life span like less than a year, should generally have people who invest with a higher amount of risk and have a single goal of creating maximum profit as possible, so general thesis also fails here, people who generally invest for a longer time span play carefully and are ought to fall under the low risk investor criteria, but still we have a similar number of investors with high risk like the other time spans which can also signal the lack of monitoring on investments because as per the project logic people who don't monitor are generally risky investors so that can be something worth noticing and needs further research that people over time fail to monitor investments and just try to complete the goal lifespan they have set in their heads before they start investing, which also counters an important factor that tells us there is no traditional benchmark of banking and investments which vary by various diverse factors and settling with a persistent risk classifier is not always optimal.

Insights:

1) The dataset was filled with people who followed a conservative behavior model, which could be proved by the analysis that around 60% of the people belonged to the low-risk category of investors, which shows the dynamic depth of the market and showed its effect on the investors.

2) The risk appetite score proved to be the most influential factor amongst our developed variables, as it had the direct impact on the risk scores that we created which was beneficial in assigning risk categories over the investors.

3) Clustering was the machine learning algorithm that was used in order to produce outputs over the dataset of over 10000 people, K-Means clustering proved to be the most successful of the clustering algorithm which grouped similar investors having similar properties together and then were examine based on the features that were created.

4) Low risk investors dominated the dataset with the count of 6828, medium risk investors had a count of 3831 and were second fiddle with respect to our dataset, High risk investors had the least count of 1341which clearly showed the dominance of conservative behavior over risk taking with respect to investing.

5) Wealth creation was the main purpose of the people investing and persisted mostly of people who belong to the working category of the age 22-30, with men dominating the count of investors.

6) Men generally showed high rise in investing from the age 26-30, while women showed early interest in investing due to the demographics of the country we live in.

7) The dataset somewhere didn't strongly align with the general speculations about investing where we had low risk investors dominating everywhere, but the ratio of high to low-risk investors didn't bother to fluctuate, which proved high uncertainty or vulnerability in the claimed assumptions.

6) Investor Behavioral Recommendation System:

A recommendation system was made using their scores in all the 4 derived metrics, the optimal score was then assigned to each which were said to be the benchmark values. The investor score which showed heavy deviation from the benchmark values were given recommendation based on the heaviest deviation. Benchmark values were generated or derived using the median scores from all the 4 derived variables, which eventually helped to validate the significance of deviation of a particular record and then label the exact needed recommendation. Suppose for example the median for all the 4 variables are as follows

Risk_Apetite:3,

Diversification_Index:2,

Stability_Index:3,

Monitoring_Frequency:3,

and, we have a record whose scores are

Risk_Apetite:4,

Diversification_Index:2,

Stability_Index:3,

Monitoring_Frequency:3,

Here, Risk Apetite is slightly greater than the median score so it would throw a recommendation like "Reduce excessive risk exposure and maintain portfolio balance." Similarly, if we have scores which are “less” as compared to the Diversification_Index, Stability_Index, Monitoring_Frequency median score’s it will throw recommendations like

"Increase portfolio diversification."

"Include more stable investment instruments."

"Monitor investments more frequently."

As visible in the logic that is presented only risk appetite throws a recommendation when the risk apetite score is greater for a particular record, other 3 variables throw a recommendation when respective scores are less for a particular record with respect to the median calculated for the variables. Our dataset consisted of a total of 12000 investors and giving" a particular suggestion to each investor was tough, so sticking to the median was of great importance and help as it gave us an idea what is the mindset and where majority of people in the bunch lie. The sole reason why median was selected over mean because 12000 records mean that we may have extensively high outliers and outliers influence mean to an extent so to be on the receiving end errorless

median was selected. As already explored, medium risk investors have a majority with high risk investors following and low risk investors being the least, it was easily understood that the median will fall somewhere between medium risk and high-risk investors.

Table 6: Recommendation count

Recommendations	Count
Current investment behaviour is well-balanced. Continue maintaining the existing investment strategy while periodically reviewing financial goals.	3664
Monitor investments more frequently.	1808
Include more stable investment instruments.	1777
Reduce excessive risk exposure and maintain portfolio balance. | Include more stable investment instruments.	1625
Reduce excessive risk exposure and maintain portfolio balance.	948
Include more stable investment instruments. | Monitor investments more frequently.	938
Reduce excessive risk exposure and maintain portfolio balance. | Include more stable investment instruments. | Monitor investments more frequently.	768
Reduce excessive risk exposure and maintain portfolio balance.	472

All the investors which closely relate or fall right on the median were labelled as "Current investment behaviour is well-balanced. Continue maintaining the existing investment strategy while periodically reviewing financial goals." So a valuable insight which suggested that almost 27% of the total investors fell closely near the benchmark that we had previously set and require no recommendations as such. Secondly, Monitoring Frequency was the second most seen recommendation with a count of 1808 which suggested people to increase their monitoring followed by stability index, Risk Appetite and Stability index and more such recommendations amongst the noticeable things were some investors even had two some even were given 3 recommendations by our system which also was possible because of the threshold we had set of the median values lying under all the 4 variables.

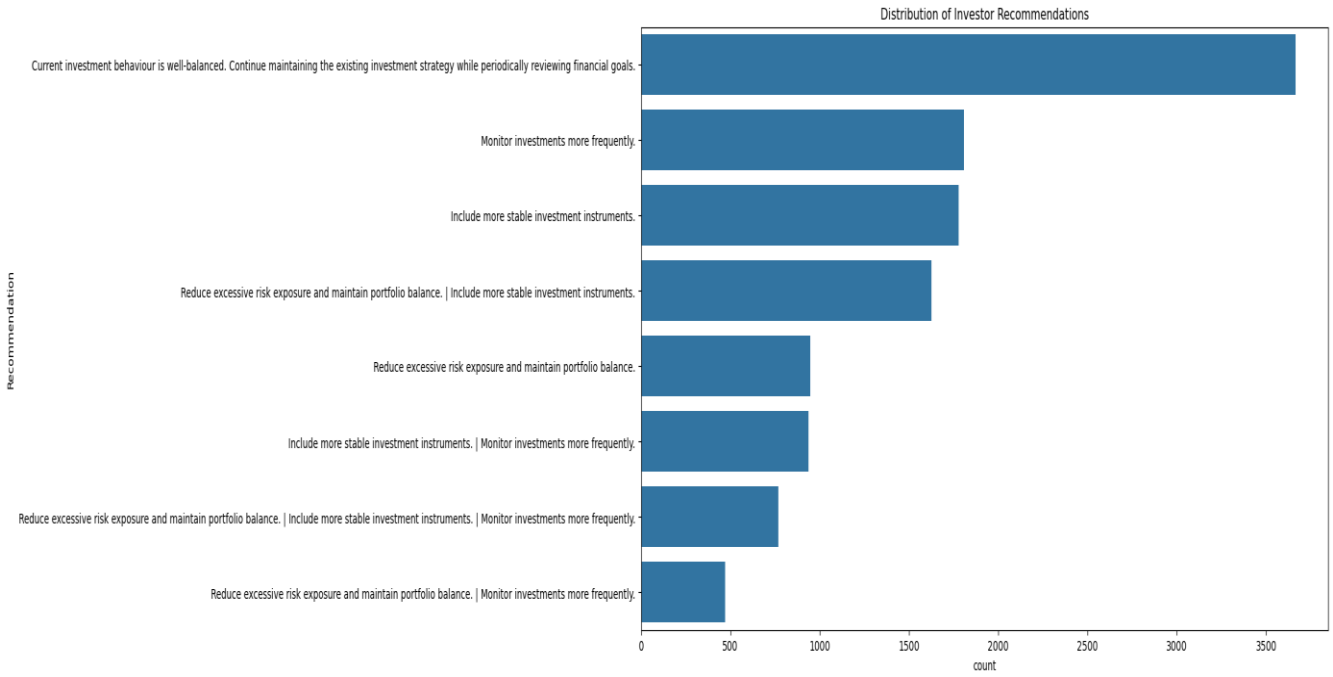


Figure 15: Distribution of recommendation count

Before creating the recommendation system the dataset already had a well computed or researched risk labels assigned to each of the 12000 records present. So after creating the recommendation system it was important to corelate and look as to which recommendation was assigned to which data risk label subordinately.

Recommendation	Current investment behaviour is well-balanced. Continue maintaining the existing investment strategy while periodically reviewing financial goals.	Include more stable investment instruments.	Include more stable investment instruments. | Monitor investments more frequently.	Monitor investments more frequently.	Reduce excessive risk exposure and maintain portfolio balance.	Reduce excessive risk exposure and maintain portfolio balance. | Include more stable investment instruments.	Reduce excessive risk exposure and maintain portfolio balance. | Include more stable investment instruments. | Monitor investments more frequently.	Reduce excessive risk exposure and maintain portfolio balance. | Monitor investments more frequently.
Risk_Category								
High Risk	1853	890	0	0	0	714	0	0
Low Risk	0	0	0	0	948	196	99	472
Moderate Risk	1811	887	938	1808	0	715	669	0

Figure 16: Recommendation Vs Risk Count

To each of the 8 recommendations a trend is seen where high risk investors either get a balanced recommendation or are recommended to take less risks, which is quite alarming that the recommendation system is working well, Secondly, despite of a few investors being low risk usually one or more than one risk variables are off the densely populated spot and may trigger recommendations that are like reducing risk increasing stability or monitoring even of it being low risk as such. As seen or explored before moderate risk investments are clustered 35% mainly towards the median and the fact that it's the most in number we get an idea of it being scattered between recommendations as well with majority being in the balanced investments closer to the mean, apart from that monitor investments frequently that is monitoring frequency is majorly the variable that affect's moderate risk investors the most.

7) Limitations:

7.1) Dataset:

The dataset consisted of 12000 records, the dataset also consisted of 24 columns which consisted of basic information as well as highlighted the interests of a person while investing, despite of it being so rich in dataset and records it lacked the general perspective thinking of an individual, so there was a need of creating more variables that focused the collective idea of the dataset columns and presented it as a whole in one column, majorly, the new derived columns were numeric and consisted the sum of all investments of a single individual as a whole.

7.2) Limited flow of algorithms:

Only 3 clustering algorithms were possible to be performed on the dataset in order limit complexity, although despite of 3 algorithms one eventually outperformed the rest which clearly showed a gap in algorithm selection, even though the best algorithm was selected to fulfill our demand

7.3) Geographical Limitation:

Even though having a large number of 12000 records in our dataset it didn't access the geographical constraint of a particular entity thus resulting in a demographic gap in the dataset which eventually resulted in analysis having a lesser scope , additionally lesser geographical scope also meant less exposure to different investing assets, which meant the study can be limited to a generic view of investors and cannot be associated to a particular group of investors.

8) Drawbacks:

8.1) Assigning risk scores:

Risk scores were assigned manually keeping in mind the importance of a particular variable in the context of risk taking, an individual may have different views regarding a particular topic and also regarding the weights assigned to the variable as such.

8.2) Limited investment variables:

To avoid complexity, also keeping in mind the limitations of variable scope and the limited records available the feature engineering was limited to 4 variables which was further used to achieve the goal of the project.

8.3) Recommendation Drawback:

The additional recommendations block can be said as a tentative behavioral improvement system that may vary depending upon various datasets and doesn't guarantee financial upbringing and only tentative on the go solutions.

9) Conclusion and Future Scope:

The study focused on applying machine learning techniques to simplify investors with respect to their risk-taking behavior, studying across 12000 investors in our dataset we were able to study the behavioral dynamics of every individual in our dataset. The project moved forward with the help of four behavioral indicators namely Risk Appetite, Stability Index, Diversification Index, Monitoring Frequency were prepared with respect to other pre-present data columns available which studied risk taking, balanced investments, understanding of various financial products, being up to date with respective financial investments were the phenomena's that each derived variable studied respectively. When deriving variables is the thing required data preprocessing and feature engineering usually comes of use which basically makes the dataset meaningful with context to machine learning approaches, deriving meaning-full variables from existing ones that can be used as an asset for the machine learning task is what feature engineering means, with preprocessing being in the upfront of feature engineering as it makes the data suitable for feature engineering that is non noisy and worthy of features being created and analysis done with the help of those features

Clustering algorithms were in action, they were used for analysis to the task of analyzing voters and segregating them into categories, as the normal parameters usually are investors were segregated in three risk categories: Low Risk Investor's, High Risk Investors, Medium Risk Investors, for that very task K-Means clustering was the algorithm used which groups related data points with one another in clusters, where we choose the "K" that is the number of clusters. K-Means analysis revealed that low risk investors form the bulk amount of the investors present in the dataset, which glorified the conservative mindset of the people and a low-risk pattern. In today's day and age investors generally invest for the purpose of creating wealth and prioritizing savings, the analysis revealed that these 2 motives out number other motives, on the contrary wealth creation and low risk investment generally don't pair up well as per traditional basics of investing, but the matchup between the two showed the balance between creating money but also being conservative or cautious about possible losses being faced. As a whole there wasn't any investing pattern that showed up as the clear winner and hence, we conclude that investment behavior majorly doesn't show any influence to any behavioral dynamics as such, with no proper way to categorize investors as low, medium, high-risk takers.

The recommendation system created identified behavioral flaws and hence suggested recommendations based on weaknesses within the variables created and suggested improvements in investing pattern to each of the tentative 12000 investors, overall the recommendation system worked well on a general platform of suggesting fixed number of recommendations within the pre prepared dataset, the recommendation system seemed to be working on the principle of median, where mostly we get everything right as per the distance of the particular point with respect to median, though based on the amount of weight given to each categorized framework where the system fails in quality recommendations as even if one defined variable is out of scope it suggests a low risk taking suggestion even to a low risk investor which could be considered as a drawback

and signals a mere use for basic suggestions of the system as here too we don’t get rid of the clear pattern to be used for a strong or a perfect investing portfolio or behavior.

10). REFERENCES:

1) Kattoura, Salem. *Investor risk profiling: an enhanced behavioral finance perspective and technique for identifying client risk profiles*. University of Wales Trinity Saint David (United Kingdom), 2022.
2) Ishii, Shogo, et al. *Capital account liberalization and financial sector stability*. Vol. 211. Washington, DC: International Monetary Fund, 200
3) Kimball, Miles S., Claudia R. Sahm, and Matthew D. Shapiro. "Imputing risk tolerance from survey responses." *Journal of the American statistical Association* 103.483 (2008): 1028-1038.
4) Nayak, Saugat. "Developing predictive models for financial stability: Integrating behavioral analytics into credit risk management." *Journal of Artificial Intelligence & Cloud Computing* 3.5 (2024): 2-10.
5) Varma, Santhosh Chitraju Gopal. "Artificial Intelligence for Behavioral Risk Analysis in Finance." (2025).
6) Qadri, Hussain Mohi-ud-Din, and M. Ishaq Bhatti. "Islamic Finance in the Modern Era." *Retrieved January* (2025).
7) Suresh, M., et al. "Analyse customer behaviour and sentiment using natural language processing (NLP) techniques to improve customer service and personalize banking experiences." *Educational Administration: Theory And Practice* 30.5 (2024): 8802-8813.
8) Dhanushkodi, Kavitha, et al. "Customer behavior analysis and predictive modeling in supermarket retail: A comprehensive data mining approach." *Ieee Access* 13 (2024): 2945-2957.
9) Kala, N. "COMBATING INSIDER THREAT USING'USER BEHAVIOURAL ANALYTICS'."
10) Ogunbodede, Olajide Olawale, et al. "Insider Threat Research: A Review of Deep Learning Approach."
11) Ogunbodede, O., et al. "Insider threat detection techniques: Review of user behavior analytics approach." *International Journal of Research in Engineering and Science* 12.9 (2024): 109-117.
12) Adeduro, Oladapo, Olabisi Josh-Falade, and Ayobami Mesioye. "Proactive Insider Threat Detection Framework: An Explainable AI and Behavioral Analytics-Driven Approach." *Journal of Future Artificial Intelligence and Technologies* 2.4 (2026): 680-697.
13) Lajuni, Nelson, et al. "The role of educators and the influence of financial behavior on personal financial distress among undergrad students of public universities in Sabah, Malaysia." *International Journal of Education, Psychology and Counseling* 2.06 (2017): 121-130.
14) Lajuni, Nelson, et al. "Religiosity, financial knowledge, and financial behavior influence on personal financial distress among millennial generation." *Jurnal Manajemen dan Kewirausahaan* 20.2 (2018): 92-98.
15) Ramalho, Thiago Borges, and Denis Forte. "Financial literacy in Brazil–do knowledge and self-confidence relate with behavior?." *RAUSP Management Journal* 54.1 (2019): 77-95.

16) Tang, Ning, and Andrew Baker. "Self-esteem, financial knowledge and financial behavior." *Journal of economic psychology* 54 (2016): 164-176.
17) Mankuroane, Tlotlego Evadia. *Analysing the factors that influence investment intentions in South Africa*. Diss. North-West University (South Africa), 2021.
18) Irandoust, Manuchehr. "Factors Associated with Financial Risk Tolerance Based on Proportional Odds Model: Evidence from Sweden." *Journal of Financial Counseling and Planning* 28.1 (2017): 155-164.
19) Ahmad, Gatot Nazir, Ari Warokka, and Irna Puji Lestari. "Financial risk tolerance analysis of Indonesian retail investors." *Humanities & Social Sciences Reviews* 8.4 (2020): 852-875.
20) Grable, John E., Abed Rabbani, and Liana Holanda Nepomuceno Nobre. "Comfort with risk: a new perspective on risk tolerance." *Research Handbook on Personal Finance* (2026): 245-264.
21) Yiu, Chi Shing, Kevin Grant, and David Edgar. "Factors affecting the adoption of Internet Banking in Hong Kong—implications for the banking sector." *International journal of information management* 27.5 (2007): 336-351.
22) Polasik, Michal, and Tomasz Piotr Wisniewski. "Empirical analysis of internet banking adoption in Poland." *International Journal of bank marketing* 27.1 (2009): 32-52.
23) Johnson, R. Burke, and Anthony J. Onwuegbuzie. "Mixed methods research: A research paradigm whose time has come." *Educational researcher* 33.7 (2004): 14-26.
24) Faber, Ronald J., and Thomas C. O'Guinn. "Classifying compulsive consumers: Advances in the development of a diagnostic tool." *Advances in consumer research* 16.1 (1989).
25) Faber, Ronald J. "Money changes everything: Compulsive buying from a biopsychosocial perspective." *American Behavioral Scientist* 35.6 (1992): 809-819.